

SYSTEM REQUIREMENTS SPECIFICATION(SRS)

This is a console-based java application booking system designed to reserve spots for users to charge their electric vehicle at a charging station. Currently, it is designed for 2 different users, which are Customers and Administrators.

The system allows customers to book a slot at the charging station, cancel the booking and get a refund, top-up their wallet account to pay for the booking made and view the history of their booking.

The system allows admins to change the status of the slots availability and also generate a report for every slot booked by the customers.

1. Slot Booking

Purpose: Enables customers to book available slots and pay using their wallet balance.

Related Tasks:

- Display a list of available slots.
- Allow users to input the preferred date of the booking.
- Users choose between 4 available slots and input the preferred time of the booking.
- Check whether users have enough funds in their wallet for the booking
- Deduct the amount from the wallet of the price for booking.

2. Booking Cancellation

Purpose: Allows customers to cancel previously booked slots.

Related Tasks:

- Prompt users to input date and time of booking they want to cancel.
- Refund the booking amount to the customer's wallet.

3. Wallet Top-Up System

Purpose: Provides customers with a way to add funds to their wallet.

Related Tasks:

- Allow customers to enter the top-up amount.
- Validate the amount and update the wallet balance.
- Save the updated balance into the system.

4. Booking History Viewing

Purpose: Displays all past and current bookings of the customer.

Related Tasks:

- Retrieve and display booking records for the logged-in customer.
- Include slot number, date and time of booking.

5. Admin Slot Management

Purpose: Gives administrators control over slot availability.

Related Tasks:

- View all slots and their statuses.
- Change the status of any slot (e.g., from available to unavailable or vice versa).

6. Generate Report for Admin

Purpose: Display all the bookings done by customers and slot availability.

Related Tasks:

- Retrieve and display booking records of all customers.
- Include name of customer, slot number, date and time of booking.
- Retrieve slot status

Assumptions:

1. The system only supports one charging station.
2. Every booking is 2 hours long.
3. Every booking cost RM10.
4. The station will only have 4 slots.
5. The maximum amount of credit customers can top up will be depended on their individual bank account.
6. Wallet top-up can only be done through banking.
7. The slots and their availability will be set up in advance by the administrators.